

Knowledge Control for Responsible Generative AI

Post-training methods to selectively **remove**, **update**, and **steer** the behavior of LLMs, multimodal LLMs, and text-to-image models — improving safety, privacy, and fairness without full retraining.



 July 2, 2026
Half-day tutorial

 2:00 – 5:30 pm
Afternoon session

 San Diego, USA
Room: Harbor D – F

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Contact

About this Tutorial

Goal & scope

Controlling the knowledge and behavior of generative AI systems — LLMs, multimodal LLMs (MLLMs), and text-to-image (T2I) models — is critical as they reach safety-sensitive, socially impactful applications. These models often encode **unintended, biased, or private** content, causing harmful outputs. **Post-training knowledge control** selectively modifies or removes model behaviors **without full retraining** — a scalable, interpretable path to safety, privacy, and fairness.

This tutorial covers the foundations and frontier methods of post-training knowledge control, bridging academia and industry: **(i)** motivations and failure modes; **(ii)** core methods — machine unlearning, knowledge editing, and inference-time interventions; and **(iii)** evaluation protocols balancing forgetting, retention, and fairness, with case studies across text and vision-language generation.

Program

Half-day · 7 parts + industry panel

I

Introduction

- What “knowledge control” means in generative models
- Motivations & taxonomy of post-training interventions
- Failure modes: harmful generation, bias, privacy leakage

20 min

II

Machine Unlearning for Knowledge Removal

- Selective forgetting & behavior suppression strategies
- Balancing stability, retention, and scalability
- Practical implications for real-world deployment

30 min

III

Knowledge Editing for Targeted Modification

- Mechanistic & memory-based editing methods
- Factual correction, personalization, reasoning consistency
- Extensions to multilingual & multimodal systems

30 min



Coffee Break

30 min

IV

Industry Perspectives on Trustworthy Control

- Deployment constraints & safety/utility trade-offs at scale
- Auditability, governance, and operational best practices
- Industry case studies & live panel discussion

PANEL35 min

V

Reflective & Bias-Aware Knowledge Control

- Bias-aware decoding & fairness-preserving interventions
- Reflection-based & self-alignment strategies
- Distillation & feedback-loop based modification

30 min

VI

Evaluation & Responsible Deployment

- Forgetting–retention & fairness–utility metrics
- Benchmarks for safety, privacy, and factual reliability
- Responsible deployment, auditing, and governance

20 min

VII

Conclusion & Future Directions

- Scalable & certifiable unlearning for foundation models
- Open challenges & cross-sector collaboration roadmap

15 min

Speakers

Academic presenters



Zheyuan Liu
Notre Dame



Yixin Wan
UCLA



Kai-Wei Chang
UCLA



Meng Jiang
Notre Dame



Jieyu Zhao
USC



Jiachen Gu
UCLA

Industry Panel

Perspectives from industry & labs



Nouha Dziri
Cohere Labs



Yuning Mao
Meta Superintelligence Labs



Jindong Gu
Google DeepMind